Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/30/2004
PI Name:	Buckey, Jay C. M.D.		
Project Title:	Otoacoustic Hearing Assessment of Space Station Crews		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Lebanon	**State:**	NH
Zip Code:	03756-0001	**Congressional District:**	2
Comments:	Address updated 9/2008		
Project Type:	Flight	**Solicitation / Funding Source:**	SMO 006
Start Date:	06/09/2000	**End Date:**	05/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:	Shuttle/ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kline-Schoder, Robert (Creare, Inc.)		
Grant/Contract No.:	NAG9-1387		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Hearing tests from the Salyut, Mir and Shuttle-Mir programs have shown temporary and, in some cases, permanent hearing threshold shifts after long-duration space flight. Currently, some areas on the International Space Station (ISS) do not meet noise standards. As a result, hearing conservation countermeasures have been implemented inflight for the astronauts. A missing component in the hearing conservation program is a reliable way to test hearing in-flight. Current methods for testing hearing while inflight have been shown in the past to be impractical due to high background noise levels. The purpose of this experiment is to develop and verify new hardware and protocols to assess and monitor hearing inflight using both distortion product otoacoustic emission (DPOAE) and audiometric hearing evaluations. Ground testing will be conducted to determine (1) the required DPOAE testing protocols for monitoring hearing in noisy environments, (2) whether background noise, like that present on ISS, renders pure-tone audiometry ineffective and (3) the feasibility of making reflectance measurements to record changes in the middle ear which can affect DPOAE and		

	audiometric measurements. Develop objective, sensitive, noise-tolerant hearing testing methods that can be used in space. Since astronauts work in a noisy environment, hearing testing is needed to insure that their hearing is not being damaged.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Noise-induced hearing loss is a significant problem in a variety of occupational settings and with aging. Reliable, noise-tolerant hearing testing methodologies could be used to assess hearing in these populations.
Task Progress:	The goal of this project “Otoacoustic Hearing Assessment of Space Station Crews” is to perform ground testing to establish the ability to record distortion product otoacoustic emissions (DPOAEs) reliably in noise. If noise tolerant protocols could be developed and if they were repeatable in noise, then they could be used on the ISS to monitor hearing. The major accomplishments of the effort have been: 1. Optimized DPOAE testing protocols to increase otoacoustic signal-to-noise ratio in noise 2. Established that otoacoustic emissions testing can be recorded in space station noise using passive hearing protection alone. 3. Established a testing protocol for determining the repeatability of otoacoustic measurements in noise. 4. Performed testing on repeatability in noise in two different settings in a total of 36 normal hearing subjects and 8 individuals with mild hearing loss. 5. Assessed the effects of postural changes on otoacoustic emissions. 6. Performed testing using otoacoustic emissions in two operational environments (KC-135 aircraft and NEEMO underwater habitat). 7. Participated in a NASA HQ-required review of all the experimental results and findings. The results of this review led to the project being accepted into the definition phase for flight on the ISS. After a successful definition phase the project has been selected to advance to the design and development phase for flight. 8. Designed, fabricated, and tested prototype hardware and software for testing otoacoustic emissions and Békésy audiometry. 9. Developed a hardware design in preparation for space qualification.
Bibliography Type:	Description: (Last Updated: 05/15/2026)
Abstracts for Journals and Proceedings	Buckey JC, Alvarenga DL, Crabtree B, Abel SM, Musiek F. "Using Otoacoustic Emissions to Assess Cochlear Function in Noise." Noise Con 2003, 2003 National Conference on Noise Control Engineering, Cleveland, Ohio, June 23-25, 2003. Noise Con 2003, 2003 National Conference on Noise Control Engineering, Cleveland, Ohio, June 23-25, 2003. , Jun-2003
Abstracts for Journals and Proceedings	Buckey JC, Alvarenga DL, Abel SM. "Reliability of Distortion Product Otoacoustic Emission Measurements in Noise." Aerospace Medical Association annual meeting 2003, San Antonio, TX, May 2003. Aviation Space and Environmental Medicine. 2003 Apr;74(4). Abstract issue. , Apr-2003
Articles in Peer-reviewed Journals	Abel SM, Crabtree B, Baranski JV, Smith DG, Thompson MM, Steeneken HJ, Verhave JA, Buckey JC, Alvarenga DL, Comtois JM. "Hearing and performance during a 70-h exposure to noise simulating the space station environment." Aviation Space Environmental Medicine 2004 Sep;75(9):764-70. PubMed PMID: 15460627 , Sep-2004